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EXPERIMENT-10

Aim : To learn Dockerfile instructions, build an image for a sample web application using DOCKERFILE

Theory :

A Dockerfile is a text file that contains a set of instructions used to automate the process of building Docker images. It defines the environment, dependencies, and commands required to run an application inside a container. Docker uses Dockerfiles to ensure consistency, portability, and reproducibility of applications across different systems.

A Dockerfile consists of several important instructions. The FROM instruction specifies the base image (e.g., Ubuntu, Node.js), while WORKDIR sets the working directory inside the container. The COPY or ADD instructions are used to transfer application files into the container. The RUN instruction executes commands during the image build process, such as installing dependencies. The CMD or ENTRYPOINT instruction defines the default command to run the application when the container starts. Additionally, EXPOSE is used to specify the port on which the application will run.

By writing a Dockerfile for a sample web application, developers can build a custom image using the docker build command and run it as a container using docker run. This approach simplifies deployment, ensures environment consistency, and allows the application to run seamlessly across development, testing, and production environments. Docker-based workflows significantly improve scalability and efficiency in modern DevOps practices.

```
 => => unpacking to docker.io/library/sample-flask-app:v1                                                    0.2s
PS C:\Users\HP 15L\Desktop\sample-flask-app> docker images
                                                                                          i Info →   U  In Use
IMAGE                 ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest    ef54e839ef54       25.9kB         9.52kB    U
sample-flask-app:v1   fde3011a5873        212MB         51.9MB
PS C:\Users\HP 15L\Desktop\sample-flask-app> docker run -d -p 5000:5000 sample-flask-app:v1
ca6680d44d322caece36b17543cc949e02b0818cb211befe6a7035baba129f2b
PS C:\Users\HP 15L\Desktop\sample-flask-app> docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS          PORTS                                       N
AMES
ca6680d44d32   sample-flask-app:v1   "flask run --host=0.…"   41 seconds ago   Up 41 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   s
ad_elion
PS C:\Users\HP 15L\Desktop\sample-flask-app>
```

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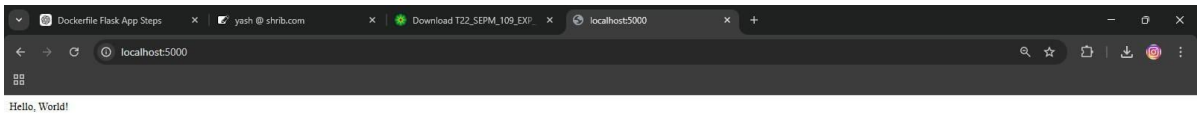
```
PS C:\Users\HP 15L\Desktop\sample-flask-app> docker --version
Docker version 29.2.1, build a5c7197
PS C:\Users\HP 15L\Desktop\sample-flask-app> docker build -t sample-flask-app:v1 .
[+] Building 12.3s (9/9) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                  0.1s
 => => transferring dockerfile: 202B                                                                  0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                   1.0s
 => [internal] load .dockerignore                                                                     0.1s
 => => transferring context: 50B                                                                      0.0s
 => [1/4] FROM docker.io/library/python:3.11-slim@sha256:fba6f3b73795df99960f4269b297420bdbe01a8631fc31ea3f121f2486d332d0  6.4s
 => => resolve docker.io/library/python:3.11-slim@sha256:fba6f3b73795df99960f4269b297420bdbe01a8631fc31ea3f121f2486d332d0  0.1s
 => => sha256:269d3f7471e27a9c2542916a49849e76630f22709b7e6063730b617d34d44d6f 14.36MB / 14.36MB      5.6s
 => => sha256:13159fd0b0512a3ecefe5d5e51affb0ef7eb36b371459c75e34f5c090a0870f4 1.29MB / 1.29MB        1.4s
 => => sha256:28c7e2bc4784ae35d32ed16d30b72e35df3c3f6a0214492f5be2b11e4b5ae2b0 250B / 250B            1.1s
 => => sha256:206356c42440674ecbdf1070cf70ce8ef7885ac2e5c56f1ecf800b758f6b0419 29.78MB / 29.78MB      5.2s
 => => extracting sha256:206356c42440674ecbdf1070cf70ce8ef7885ac2e5c56f1ecf800b758f6b0419             0.5s
 => => extracting sha256:13159fd0b0512a3ecefe5d5e51affb0ef7eb36b371459c75e34f5c090a0870f4             0.1s
 => => extracting sha256:269d3f7471e27a9c2542916a49849e76630f22709b7e6063730b617d34d44d6f             0.3s
 => => extracting sha256:28c7e2bc4784ae35d32ed16d30b72e35df3c3f6a0214492f5be2b11e4b5ae2b0             0.0s
 => [internal] load build context                                                                     0.1s
 => => transferring context: 194B                                                                     0.0s
 => [2/4] WORKDIR /app                                                                                0.2s
 => [3/4] RUN pip install flask==2.3                                                                  2.1s
 => [4/4] COPY . /app                                                                                 0.1s
 => exporting to image                                                                                1.1s
 => => exporting layers                                                                               0.6s
 => => exporting manifest sha256:247bc7bbcf2f8acc15a922b40f48d1bbc956b2c8c5023ccf8fa0732e1d92a3b2     0.0s
 => => exporting config sha256:1ac3e6b904147304c26dcd5e51dca9853974e68393f14b02a0eae78059033e88       0.0s
 => => exporting attestation manifest sha256:b58db5a08e0bda4ebb0d353029f4ebcae4ad283149bdceb86a077cb1ce803d8a  0.1s
 => => exporting manifest list sha256:fde3011a58738f57ef957872e1598ec24530ca3c73c845f25a07bc954746919e  0.0s
 => => naming to docker.io/library/sample-flask-app:v1                                                0.0s
 => => unpacking to docker.io/library/sample-flask-app:v1                                             0.2s
```